

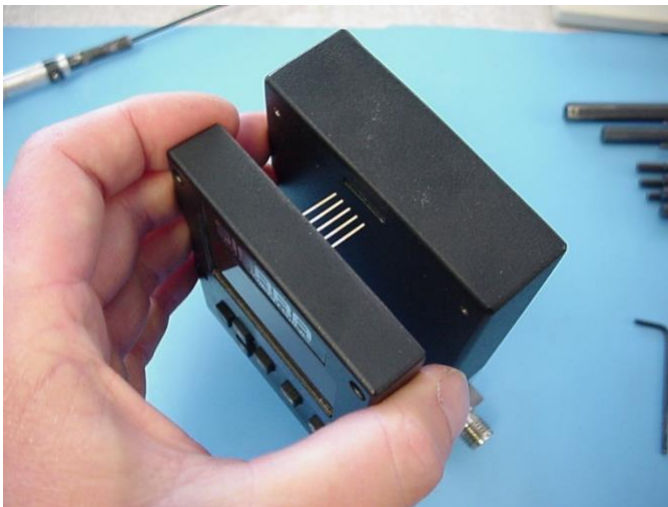
100-series controller field valve adjustment procedure

Symptom: Flow dropping off after a few minutes.

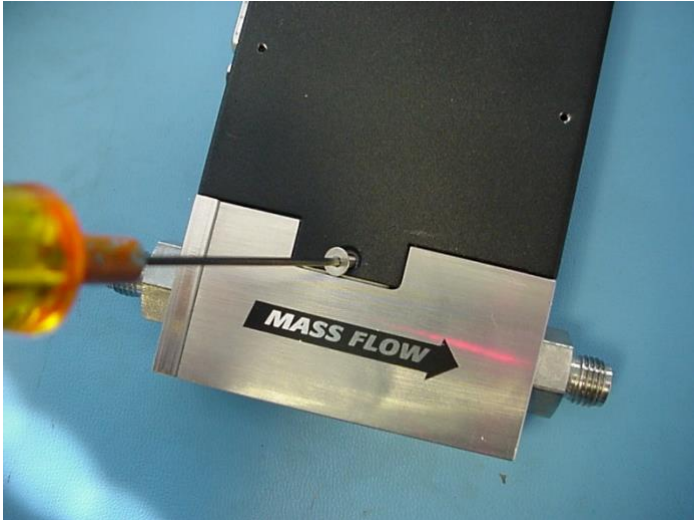
1. Turn off the power to the instrument.
2. If you do not have the Pilot Module display, move on to step 3. If you have a Pilot Module display, it must be removed to open the instrument for adjustment. To remove the Pilot Module, use an Allen wrench to unscrew the 4 screws from the front corners of the Module.



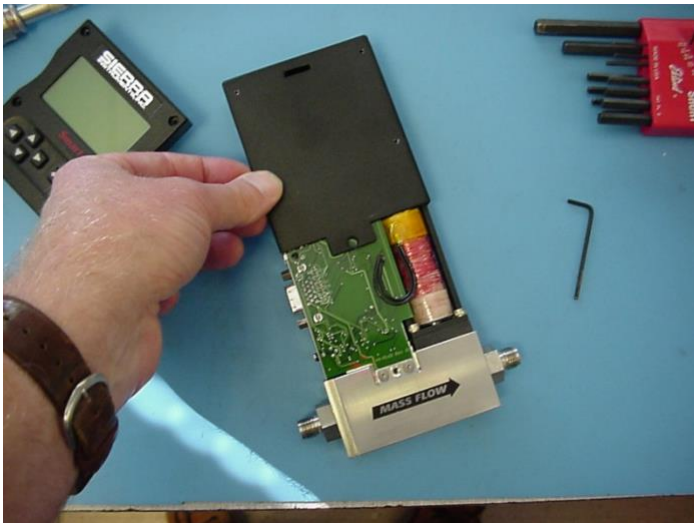
3. With the screws removed, carefully pull the Pilot Module in the direction perpendicular to the instrument. The 5 connecting pins will come free from the slot in the metal electronics enclosure and you can set the Pilot Module aside.



4. Remove the 2 screws that secure the black cover onto the steel flow body. These screws are located on the small tabs at the base of the black cover, one on the front side and one on the rear.

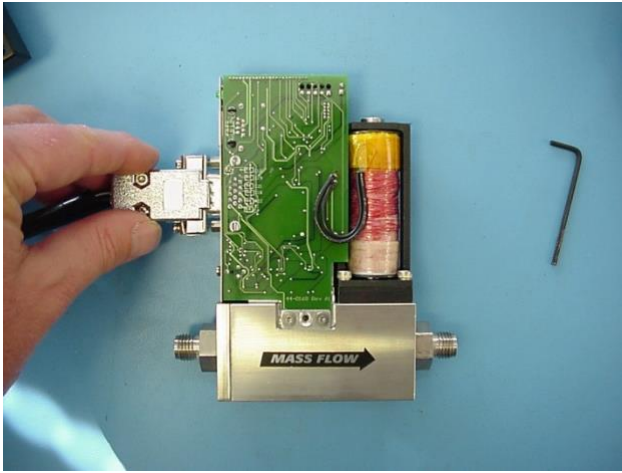


5. Slowly, slide the black cover straight up and off the instrument, then put aside.

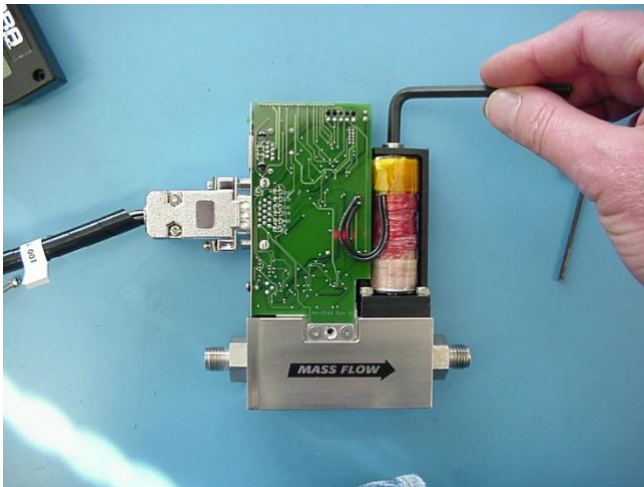


6. Find the large Allen socket on top of the valve assembly. This is at the outlet side of the controller, beside the circuit boards. Select the correct wrench to fit this socket, but do not adjust yet.

7. When you are ready, apply power and gas flow to the instrument. DO NOT TOUCH THE CIRCUIT BOARDS AT THIS TIME—THE INSTRUMENT COULD BE DAMAGED.



8. Run the instrument until the flow begins to “drop off.” Insert the Allen wrench into the valve adjustment screw and turn the screw counter-clockwise until the flow increases. A small turn (approx $\frac{1}{4}$ revolution) should be enough.



9. Set the instrument for the maximum flow rate you will require. Observe that the instrument can attain this flow rate and maintain it for several minutes.
10. When you are satisfied that the device can reach this flow value, turn off the power to the device and shut off the gas flow.
11. VERY CAREFULLY, re-install the black cover over the instrument, pulling slightly out on the tabs at the bottom (where the 2 screws attach) so that the inside of the cover does not drag on the circuit boards when you slide it into position.
12. When the cover is back in position, re-attach the 2 screws.
13. If you have no Pilot Module, you are done! If you have the Pilot Module display, re-attach it using the reverse of the procedures 2-3 above.

Symptom: Flow leaks past valve when valve is closed

Follow steps 1-7 above. When you have applied power to the instrument, make the setpoint ZERO or activate the VALVE CLOSED command.

6. Check the inlet pressure to the mass flow controller so that it matches the pressure of your process. Check the down-stream pressure from the controller so that it also matches your process. These pressures should be the same as the pressures listed on the data label and calibration certificate. If they are different, contact the factory or your Sierra representative.
7. With the power applied for a minimum of 10 minutes and your pressures correctly set, determine the amount of the gas leak (your instrument will tell you!).
8. Slowly turn the valve adjustment screw with the Allen wrench until the leak stops. Do NOT turn any further. If you need to make more than ½ turn to stop this leak, contact Sierra.
9. Next, change the setpoint to the instrument full scale flow. Confirm that the measured mass flow rate increases to the full scale value (equal to the setpoint).
If your controller reaches full scale flow, your adjustment is finished. Proceed to step 10.
If the controller stops before it reaches the full scale value, you may have adjusted the valve too tight. Change the setpoint back to ZERO and loosen the valve adjustment screw until the leak returns. Repeat steps 7-8. Change the set point to full scale again.
If your controller stops at a flow value below full scale your valve may be damaged. Contact Sierra for instructions.
10. VERY CAREFULLY, re-install the black cover over the instrument, pulling slightly out on the tabs at the bottom (where the 2 screws attach) so that the inside of the cover does not drag on the circuit boards when you slide it into position.
11. When the cover is back in position, re-attach the 2 screws. If you removed the Pilot Module display, reattach it by inserting the 5 connecting pins through the slot in the enclosure and then insert and tighten the 4 mounting screws.
12. Your controller is now ready to be placed back into service.

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